

Reflection in C++26

From P2996 to a 105-line Rosetta

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What is reflection?

"The ability of software to expose its internal structure."

Static reflection: the **compiler** exposes structure at compile time

P2996 — one of the **largest** proposals in C++ history.

Why should you care?

One C++ description → *everything*:

- **Automatic serialization / deserialization**
- **Generic struct printing / debugging**
- **Compile-time generation of types via splicers**
- **Doc generation** (md, html...)
- Annotations (P3394, paired with reflection) let you attach compile-time metadata to declarations and read it back via reflection, supporting things like validation rules or serialization hints.
- ...

The two operators in one picture

```
constexpr std::meta::info info = ^^Circle;           // reflect-on a type
typename [: info :] c = {.name = "c", .radius = 1.0}; // splice it back into code
// also works on members, expressions, namespaces, templates...
```

- Everything that crosses the boundary is `constexpr` / `constexpr`
- `^^T` lifts the *name* `T` into a `constexpr` value (`std::meta::info`)
- `[: info :]` drops a `std::meta::info` value back into a *type, expression, or member*

Why not earlier? — the design space

Every reflection system trades along the same axes:

Axis	Endpoints
When	Compile-time only ↔ Runtime-queryable
Intrusiveness	Non-intrusive ↔ Macros / annotations required
Source of metadata	Native compiler ↔ External codegen ↔ Manual
Target use	Serialization ↔ Bindings ↔ GUIs ↔ RPC

C++ historically lacked native reflection → **dozens** of libraries with different philosophies.

Why not earlier? — C++ libraries pre-P2996

Library	Type	Intrusive?	Coverage
RTTI (typeid)	language	none	type identity only
Boost.PFR	header	none	aggregates only
magic_enum	header	none	enums (via __PRETTY_FUNCTION__)
RTTR	header	macro	fields, methods, inheritance
EnTT::meta	header	registration	data + funcs (games)
Boost.Describe	header	macro	members + bases
Qt MOC	codegen	Q_OBJECT	full + signals/slots
Unreal UHT	codegen	UCLASS	full + GC + Blueprint
SWIG	codegen	none	multi-language bindings

Why not earlier? — every other language has this

Language	Runtime	Compile-time	Idiom
Python	● deep	●	<code>inspect</code> , <code>getattr</code> , <code>decorators</code>
Java	● native	● processors	<code>java.lang.reflect</code>
JavaScript	● Proxy	●	<code>Reflect</code> , <code>Object.*</code>
Rust	● minimal	● proc-macros	<code>derive</code> , <code>serde</code> , <code>bevy_reflect</code>
C++26	● build manually	● P2996	<code>^^T</code> , <code>[: r :]</code> , <code>consteval</code>

C++ was the only major language without first-class reflection — until now.

Lessons from the neighbours

- D is the spiritual ancestor of P2996 — `__traits` + `static foreach` shaped the C++26 design.
- Rust `bevy_reflect` proves runtime registries can stay idiomatic.
- Go's `struct tags` map cleanly onto P3394 annotations.
- C# `attributes` + `source generators` — cleanest runtime/codegen hybrid.

C++26 closes the `language-level` gap.

Runtime registries (Rosetta, RTTR, EnTT) don't go away — they get `auto-filled`.

Supported compilers

Compiler	Reflexion C++26	template for
GCC 16.1+	✓ Almost done	✓
Clang (Bloomberg)	✓ Almost done	✓ (<code>-freflection-latest</code>)
Clang mainline	● Partial	● In progress
MSVC	✗	✗
EDG	● Partial	●

A running example

```
struct Point {  
    int    x;  
    int    y;  
    double z;  
    double norm() const { return ...; }  
    Point scale(double) const { return ...; }  
};
```

Enumerate `Point`'s fields

```
constexpr auto ctx = std::meta::access_context::current();  
constexpr auto fields = std::define_static_array(std::meta::nonstatic_data_members_of(^Point, ctx));  
template for (constexpr auto f : fields) {  
    std::println(  
        "{} : {}",  
        std::meta::identifier_of(f),  
        std::meta::display_string_of(std::meta::type_of(f))  
    );  
}
```

Output:

```
x : int  
y : int  
z : double
```

No macros, no inheritance, no registration, non intrusive. Just the type.

Enumerate `Point`'s methods

```
constexpr auto members = std::define_static_array(std::meta::members_of(^Point, ctx));

template for (constexpr auto m : members) {
    if constexpr (is_exportable_member_function(m)) {
        std::print("  {} {}(", std::meta::display_string_of(std::meta::return_type_of(m)),
                    std::meta::identifier_of(m));

        bool first = true;
        template for (constexpr auto param: std::define_static_array(std::meta::parameters_of(m))) {
            if (!first) std::print(", ");
            first = false;
            std::print("{} ", std::meta::display_string_of(std::meta::type_of(param)));
        }
        std::println(")");
    }
}
```

Output:

```
double norm()
Point scale(double)
```

Simple JSON-ish serialization

```
template <typename T>
std::string to_json(const T& obj) {
    std::string out = "{";
    bool first = true;
    template for (constexpr auto m : std::define_static_array(
        std::meta::nonstatic_data_members_of(^T))) {
        if (!first) out += ",";
        first = false;
        out += "'";
        out += std::meta::identifier_of(m);
        out += "\"";
        out += serialize_value(obj.[:m:]); // your per-type value formatter
    }
    out += "}";
    return out;
}
```

Serialization example

```
enum class Color { Red, Green, Blue };

struct Address {
    std::string street;
    int         number;
};

struct Person {
    std::string      name;
    int              age;
    bool             active;
    Color            favorite_color;
    Address          home;
    std::vector<std::string> tags;
};

int main() {
    Person p{
        .name      = "Alice",
        .age       = 30,
        .active     = true,
        .favorite_color = Color::Green,
        .home       = {"Rue Pasteur", 42},
        .tags       = {"admin", "ops"},
    };

    std::println("{} ", to_json(p));
}
```

Serialization result

```
{  
  "name": "Alice",  
  "age": 30,  
  "active": true,  
  "favorite_color": "Green",  
  "home": {  
    "street": "Rue Pasteur",  
    "number": 42  
  },  
  "tags": [  
    "admin",  
    "ops"  
  ]  
}
```

Aggregate example

Use `std::meta::define_aggregate`:

(previously called `std::meta::define_class` in draft)

```
#include <meta>

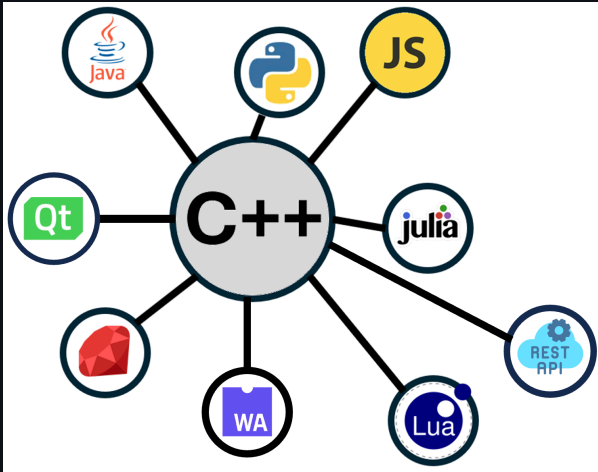
struct Synthesized; // incomplete – to be defined

consteval {
    std::meta::define_aggregate(^^Synthesized, {
        std::meta::data_member_spec(^^int,    {.name = "id"}),
        std::meta::data_member_spec(^^double, {.name = "value"}),
    });
}

// Synthesized is now: struct Synthesized { int id; double value; };
```

Etonish, nein ? 😊

The example of Rosetta



- A C++ introspection & automatic language binding.
- Non intrusif
- Introspection **at runtime**
- Hand write registration à la `pybind11`

<https://github.com/xaliphostes/rosetta>

Rosetta — *before* C++26

For each class, hand-write the registration:

```
rosetta::Class<Shape>("Shape")
    .field("name", &Shape::name).doc("display name")
    .method("describe", &Shape::describe).doc("greeting")
    .method("area", &Shape::area)
    .static_method("next_id", &Shape::next_id);

rosetta::Class<Circle>("Circle")
    .base<Shape>()
    .field("radius", &Circle::radius)
        .doc("radius in meters")
        .range(0.0, 1e6);

// ... repeat for Rectangle, every other type, every project ...
```

Real numbers from our internal lib: ~**6000** lines of glue.

Rosetta — *with* C++26

Use C++ annotations for doc, readonly, numeric range, alias, displayed-name...

```
rosetta::register_reflected<Shape>();  
rosetta::register_reflected<Circle>();  
rosetta::register_reflected<Rectangle>();
```

That's it:

- Same registry
- Same downstream generators

~100 lines of `register_reflected` machinery — once, in the library — and *you never touch it again!*

Rosetta before and after C++26 (lines of code)

Spec	Before	After
core	5,903	106
common	2,380	0
js	1,808	204
py	1,553	90
rest	3,327	305
wasm	1,435	102
Total	16,500	807

Applications — Auto language bindings

One C++ description, N hosts:

Target	What reflection drives
Python	<code>pybind11</code> <code>def(...)</code> , <code>__init__</code> , dunder methods
JavaScript	Node.js N-API wrappers, getters/setters
WebAssembly	Emscripten <code>EMSCRIPTEN_BINDINGS</code>
TypeScript	<code>.d.ts</code> declaration files
REST API	HTTP routes mapped to methods, JSON I/O
Lua / Ruby / Julia	Sol2, Rice, CxxWrap
Java/JNI, C#/.NET, Swift	bridge generation

The alternative is N hand-maintained binding files that drift as the C++ API evolves.

Applications — Validation · Docs · Persistence · Live

- **Validation** — range, regex, not-null, cross-field invariants (from annotations).
- **Documentation** — Markdown / HTML / OpenAPI / Sphinx from `doc("...")`.
- **Persistence / ORM** — `CREATE TABLE`, migrations from class diffs.
- **Configuration & DI** — config-file → object hydration, CLI flag generation.
- **Live scripting / REPL** — every method auto-callable from the console.
- **Testing** — property-based, fuzzing, snapshot tests, binding-coverage.

Applications — Serialization · GUI · REST

Serialize: JSON / XML / YAML / TOML / CBOR / MsgPack / Protobuf / HDF5 — *no schema needed.*

GUI generation:

- Qt property editors (`ObjectInspector<Circle>` for free)
- QML data binding, web forms, Blueprint-style node editors
- ImGui debug panels — sliders, color pickers, drag-floats per field
- ...

REST / RPC: each method → endpoint, each parameter → JSON field.

```
for (auto cls : registry.list_classes())  
    for (auto m : cls->methods())  
        router.add(cls->name + "/" + m.name, &dispatch);
```

Annotation is the leverage

One `[[=doc{...}, =range{...}]] double radius;` is **simultaneously**:

- A Python attribute
- A JavaScript getter/setter
- A REST endpoint
- A JSON-serializable element
- A Qt property in the inspector
- A documented entry in the reference
- A range-validated database column
- ...

Every line of registration unlocks features across multiple subsystems at once.

QML example

```
struct Person {
    [[= rosetta::doc{"the person's display name"}]]
    std::string name;

    [[ = rosetta::doc{"age in whole years"}, = rosetta::range{0.0, 150.0} ]]
    int age = 0;

    [[ = rosetta::doc{"server-assigned identifier"}, = rosetta::readonly{} ]]
    std::string id;

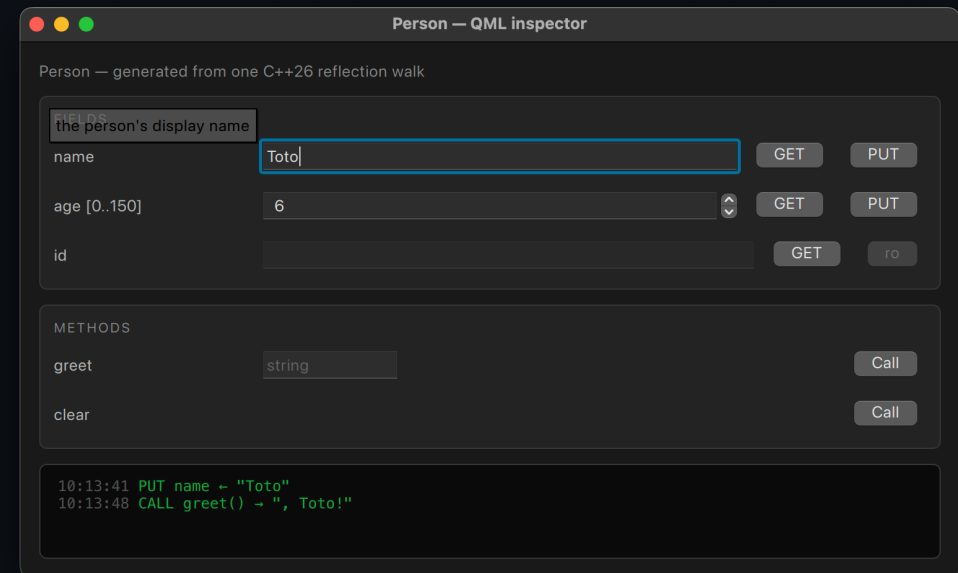
    Person() = default;
    Person(std::string n, int a, std::string i): name(std::move(n)), age(a), id(std::move(i)) {}

    [[= rosetta::doc{"Returns a greeting prefixed by the given salutation."}]]
    std::string greet(const std::string &salutation) const {
        return salutation + ", " + name + "!";
    }

    void clear() {
        name.clear();
        age = 0;
    }
};
```

QML example (suite)

```
Person person;  
rosetta::ReflectedObject reflected;  
rosetta::bind_qml<Person>(&reflected, person);  
  
QQmlApplicationEngine engine;  
engine.rootContext()->setContextProperty("inspector", &reflected);
```



When *not* to use reflection

- **Hot inner loops** — virtual dispatch and `Any` boxing have measurable cost; keep numerical kernels typed.
- **Tiny throw-away tools** — registration is overhead if no extension needs it.
- **Highly templated header-only math types** — nothing to expose.
- **When compile-time reflection alone covers your needs** — skip the runtime registry.

Use it where leverage is high: APIs that cross language boundaries, are edited live, persisted, validated, documented, versioned — **application-shaped code**, not algorithmic kernels.

References

- [Reflection proposal](#)
- [Python Bindings with Value-Based Reflection](#)
- [Using Reflection to Replace a Metalanguage for Generating JS Bindings](#)
- [Short video presentation for bindings](#)

Questions?

Example Python binding

```
template <typename T> void bind(py::module_ &m, const char *py_name) {
    py::class_<T> cls(m, py_name);
    cls.def(py::init<>());

    // Fields
    template for (constexpr auto fld : std::define_static_array(std::meta::nonstatic_data_members_of(^T, ctx))) {
        if constexpr (ro) {
            // readonly annotation -> Python read-only property
            cls.def_property_readonly(name, [] (const T &self) -> MemberT {
                return self.[:fld:]; }, docstr);
        }
    }

    // Methods
    template for (constexpr auto fn : std::define_static_array(std::meta::members_of(^T, ctx))) {
        if constexpr (is_exportable_member_function(fn)) {
            constexpr auto name = std::define_static_string(std::meta::identifier_of(fn));
            constexpr auto m_doc = std::meta::annotation_of_type<doc>(fn);
            constexpr const char *mdoc = m_doc.has_value() ? m_doc->text : "";

            if constexpr (std::meta::is_static_member(fn)) {
                cls.def_static(name, &[:fn:], mdoc);
            } else {
                cls.def(name, &[:fn:], mdoc);
            }
        }
    }
}
```